Math review.

Laplace. Transform.

$\mathcal{L}[f(t)] = \int_0^\infty f(t) e^{-st} dt$.

$f(t) = \begin{cases} 0 & t<0 \\ Ae^{-at} & t\geq 0 \end{cases} \Rightarrow F(s) = \frac{A}{s+a}$.

$f(t) = \begin{cases} 0 & t<0 \\ A & t\geq 0 \end{cases} \Rightarrow F(s) = \frac{A}{s}$.

$f(t) = \begin{cases} 0 & t<0 \\ At & t\geq 0 \end{cases} \Rightarrow F(s) = \frac{A}{s^2}$.

simusoidal

$\mathcal{L}[A\sin\omega t] = \frac{A\omega}{s^2+\omega^2}$ $\quad \mathcal{L}[A\cos\omega t] = \frac{As}{s^2+\omega^2}$.

$\delta(t) = \begin{cases} \lim_{t_0\to 0} \frac{A}{t_0} & 0<t\leq t_0 \\ 0 & \end{cases} \Rightarrow \mathcal{L}[\delta(t)] = A$

Properties

Differential theorem $\mathcal{L}[f(t)] = F(s)$

$\mathcal{L}\left[\frac{d^n f(t)}{dt^n}\right] = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \cdots$.

Final value theorem

$\lim_{t\to\infty} f(t) \Rightarrow \lim_{s\to 0} sF(s)$ $\left(\lim_{t\to\infty} f(t) \text{ exist}\right)$.

Inverse. Laplace. Transform.

$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j}\int_{\sigma-j\omega}^{\sigma+j\omega} F(s) e^{st} ds$.

case A: $F(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s+p_1)(s+p_2)\cdots} = \frac{k_1}{s+p_1} + \frac{k_2}{s+p_2} + \cdots$

其中 $k_K = (s+p_K)F(s)\big|_{s=-p_K}$.

case B $F(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s+p)^r} = \frac{k_1}{s+p} + \frac{k_2}{(s+p)^2} + \cdots + \frac{k_r}{(s+p)^r}$.

$k_r = (s+p)^r F(s)\big|_{s=-p}$.

$k_{r-1} = \frac{1}{1!}\frac{d}{ds}[(s+p)^r F(s)]\big|_{s=-p}$.

$k_1 = \frac{1}{(r-1)!}\frac{d^{r-1}}{ds^{r-1}}[(s+p)^r F(s)]\big|_{s=-p}$.

case C. $\mathcal{L}[Ae^{-at}\cos\omega t] = A\frac{s+a}{(s+a)^2+\omega^2}$.

$\mathcal{L}[Ae^{-at}\sin\omega t] = A\frac{\omega}{(s+a)^2+\omega^2}$.

Eg. $F(s) = \frac{s+2}{s^2+2s+5} = \frac{s+1}{(s+1)^2+2^2} + \frac{1}{2}\frac{2}{(s+1)^2+2^2}$

$\mathcal{L}^{-1}F(s) = e^{-t}\cos 2t + \frac{1}{2}e^{-t}\sin 2t$.

Transfer function.

$r(t) \to \boxed{G(s)} \to c(t)$

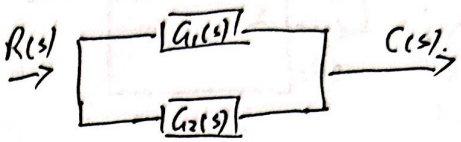
$G(s) = \frac{\mathcal{L}[\text{output}]}{\mathcal{L}[\text{input}]}$.

Series: $R(s) \to \boxed{G_1(s)} \to \boxed{G_2(s)} \to C(s)$

$\Downarrow$

$R(s) \to \boxed{G_1(s)G_2(s)} \to C(s)$

Parallel.



$G(s) = G_1(s) + G_2(s)$.

Model. { Mechanical. / Electrical. / Mechatronic.

General { Common. approach based. energy/power / Lumped parameter. / Look for 2 power variable.

Power = $f \times v = \tau \times w = u \times i$

Mechanical (translational)

Power = $f \times v$

位移 $x(t) = \int_0^t v(t)dt$ $\quad v = \frac{dx}{dt}$.

冲量 $p(t) = \int_0^t f(t)dt$ $\quad f = \frac{dp}{dt}$.

Energy

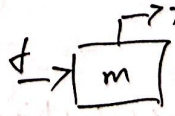
Spring $\Delta W = f(vdt) = fdx$ 势能.

Mass $\Delta W = v(fdt) = vdp$ 动能.

Damper $\Delta W = (vf)dt$ 耗散.

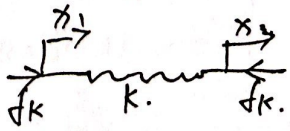
本构方程

刚体



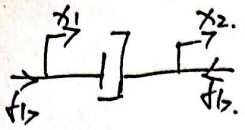
$f = m\frac{d^2x}{dt^2}$.

理想弹簧.

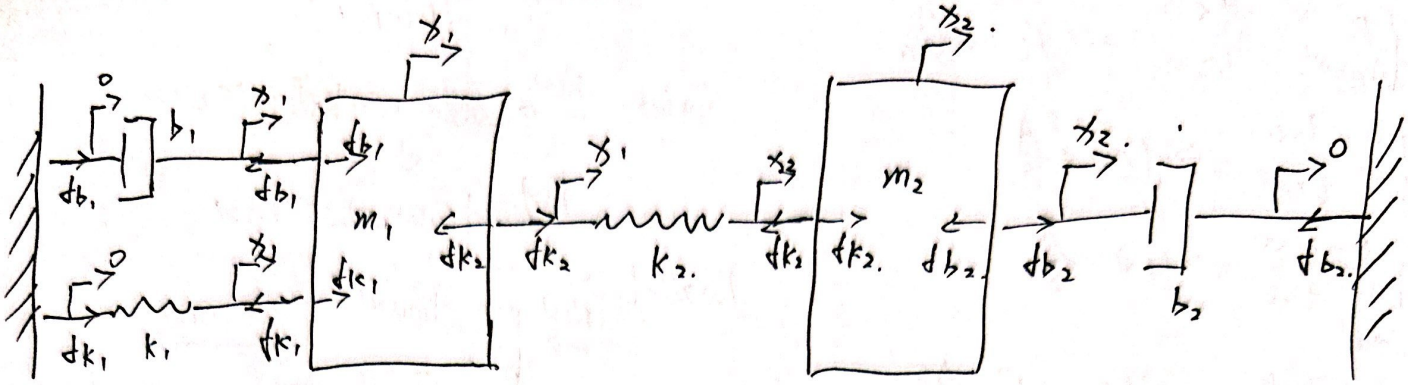


$f_k = k(x_1 - x_2)$

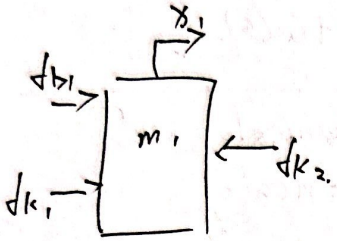
Ideal Damper.

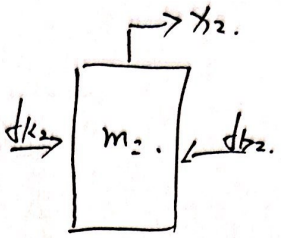


$f_b = b(\dot{x}_1 - \dot{x}_2)$



其中.





$m_1: \Sigma f = f_{b_1} + f_{k_1} - f_{k_2} = m_1 \ddot{x}_1$

$m_2: \Sigma f = f_{k_2} - f_{b_2} = m_2 \ddot{x}_2$.

$f_{b_1} = b_1(0 - \dot{x}_1) = -b\dot{x}_1$

$f_{k_1} = k_1(0 - x_1) = -k_1 x_1$

$f_{b_2} = b_2(\dot{x}_2 - 0) = b_2 \dot{x}_2$.

$f_{k_2} = k_2(x_1 - x_2)$

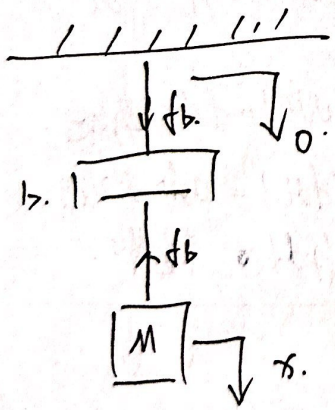
$\Rightarrow$

$$m_1 \ddot{x}_1 + b_1 \dot{x}_1 + k_1 x_1 + k_2 x_2 = 0.$$

$$m_2 \ddot{x}_2 + b_2 \dot{x}_2 - k_2 x_2 = 0.$$

Dynamic equation.

Eg. 降落伞.

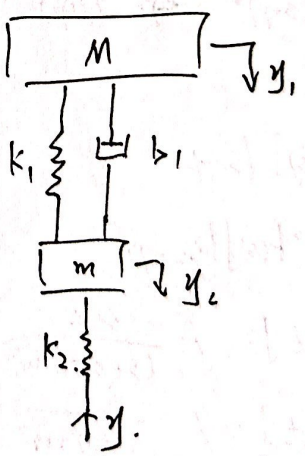


$f_b = b(0 - \dot{x})$

$f_b = m\ddot{x}$.

$\Rightarrow m\ddot{x} + b\dot{x} = 0$

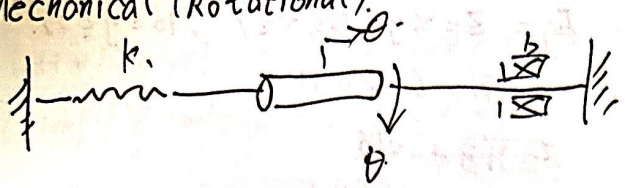
1/4 Car.



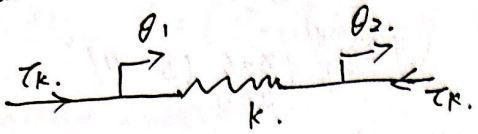
$$\frac{Y_1(s)}{Y(s)} = \frac{-k_2(b_1 s + k_1)}{(ms^2 + b_1 s + k_1 + k_2)(Ms^2 + b_1 s + k_1) - (b_1 s + k_1)^2}$$

$$\frac{Y_2(s)}{Y(s)} = \frac{-k_2(Ms^2 + b_1 s + k_1)}{(ms^2 + b_1 s + k_1 + k_2)(Ms^2 + b_1 s + k_1) - (b_1 s + k_1)^2}.$$

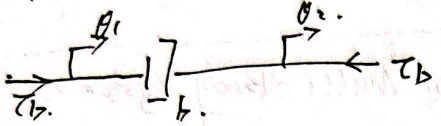
Mechanical (Rotational).



Ideal Elements.



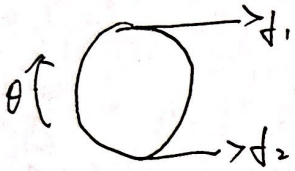
$\tau_k = k(\theta_1 - \theta_2)$.



$\tau_b = b(\dot\theta_1 - \dot\theta_2)$

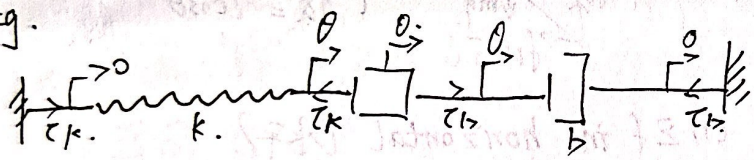


$\tau = J\ddot\theta$.



$J\ddot\theta = r(f_1 - f_2)$

Eg.

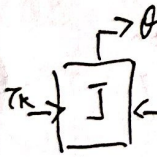


$\tau_k = -k\theta$

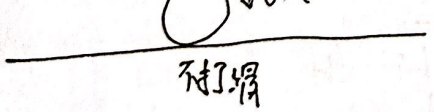
$\tau_b = b\dot\theta$.

$\tau_k - \tau_b = J\ddot\theta$

$\Rightarrow J\ddot\theta + b\dot\theta + k\theta = 0$.



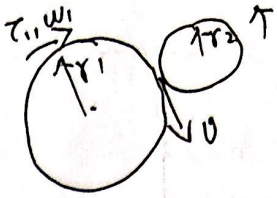
Two-port System.



$v = w r = \dot\theta r$.

$w\tau = fv$.

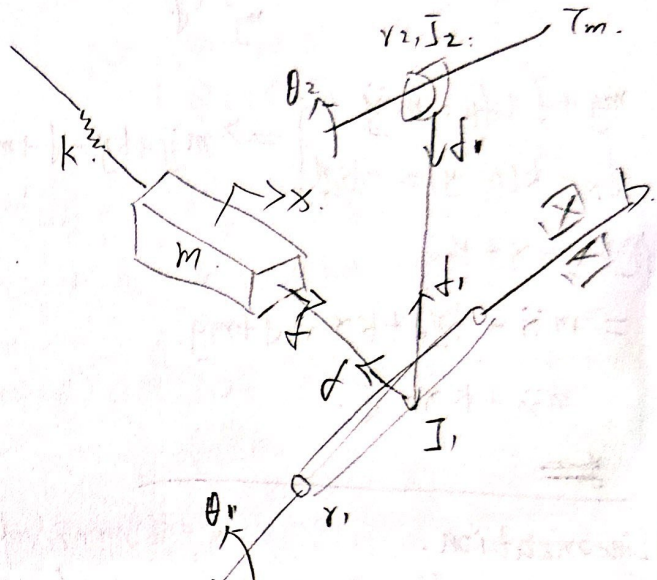
②.



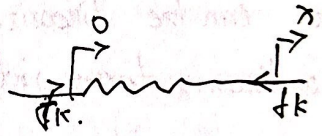
$v = w_1 r_1 = w_2 r_2$.

$w_1 \tau_1 = w_2 \tau_2$.

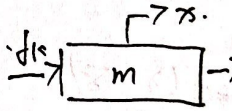
Eg.



Find $\frac{X(s)}{\tau_m(s)}$.

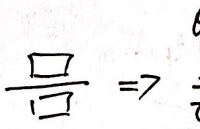


$f_{k1} = k(0 - x) = -kx$

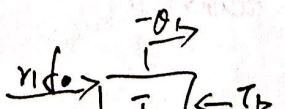


$f + f_k = m\ddot x$.

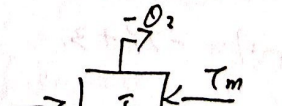
$\theta = -\theta_1$



$\Rightarrow \tau_b = -b\dot\theta_1$



$\Sigma \tau = r_1 f - r_1 f_1 - \tau_b = J_1(-\ddot\theta_1)$

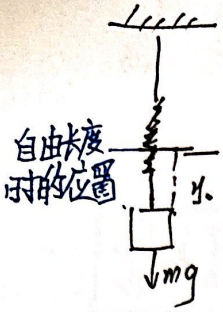


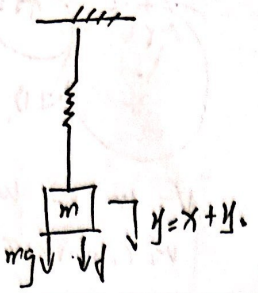
$\Sigma = r_2 f_1 - \tau_m = J_2(-\ddot\theta_2)$

$x = \theta_1 r_1 = \theta_2 r_2 \Rightarrow$

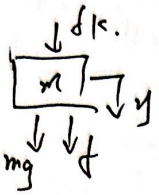
$$\left(m + \frac{J_1}{r_1^2} + \frac{J_2}{r_2^2}\right)\ddot x + \frac{b}{r_1^2}\dot x + kx = \frac{\tau_m}{r_2}$$

平衡点..





$mg = ky_0$.



$$\left.\begin{aligned} mg + f + f_k &= m\ddot{y} \\ f_k &= k(0-y) = -ky \end{aligned}\right\} \Rightarrow m\ddot{y} + ky = f + mg.$$

代入 $y = x + y_0$.

$$\Rightarrow m\ddot{x} + ky_0 + kx = f + mg.$$

$$m\ddot{x} + kx = f.$$

~~x~~

---

linearization.

For a nonlinear system $y = f(x)$ in a small region around $\bar{x}$ [平衡点 eq.], the dynomic equation can be linearized. by expanding the nonlinear terms into Tylor Series.

$$y = f(x) = f(\bar{x}) + \frac{df}{dx}\Big|_{x=\bar{x}}(x-\bar{x}) + \frac{1}{2!}\frac{d^2f}{dx^2}\Big|_{x=\bar{x}}(x-\bar{x})^2 + \cdots$$

high order terms = 0.

Eg. $f(x) = \frac{1}{x_2}$ $(\bar{x}=1)$.

$$f(x) = f(1) + \frac{d(\frac{1}{x_2})}{dx}\Big|_{x=1}(x-1)$$

$$= 1 + (-2)(x-1) = -2x+3.$$

$y = f(x_1, x_2)$

$$= f(\bar{x}_1, \bar{x}_2) + \frac{\partial f}{\partial x_1}\Big|_{\substack{x_1=\bar{x}_1 \\ x_2=\bar{x}_2}}(x_1-\bar{x}_1) + \frac{\partial f}{\partial x_2}\Big|_{\substack{x_1=\bar{x}_1 \\ x_2=\bar{x}_2}}(x_2-\bar{x}_2).$$

Eg $z = xy$, $5 \le x \le 7$, $10 \le y \le 12$.

$\bar{x} = 6$, $\bar{y} = 11$

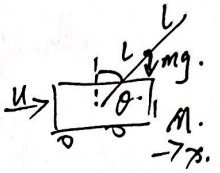
~~$z = \bar{x}\bar{y} + d(xy)$~~

$$z = \bar{x}\bar{y} + \frac{\partial(xy)}{\partial x}\Big|_{\substack{x=6 \\ y=11}}(x-6) + \frac{\partial(xy)}{\partial y}\Big|_{\substack{x=6 \\ y=11}}(y-11)$$

$$= 11x + 6y - 66$$

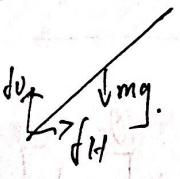
Dynamic of Multi-Body Systems.

① Newton-Euler Method.



Free body diagram:

Pendulum



$$\begin{cases} x_a = x + l\sin\theta \\ y_a = l\cos\theta \end{cases}$$

(1) Σf in horizontal (水平).

$$f_H = m\ddot{x}_a = m\frac{d}{dt}\left(\frac{dx_a}{dt}\right).$$

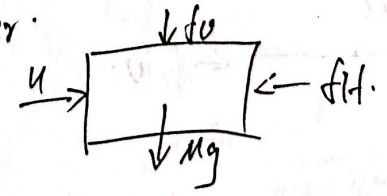
$$= m(\ddot{x} + l\ddot{\theta}\cos\theta - l\sin\theta\dot{\theta}^2).$$

(2) Σf in vertical.

$$f_v - mg = m\ddot{y}_a = -ml[\cos\theta\cdot\dot{\theta}^2 + \sin\theta\ddot{\theta}].$$

(3) $\Sigma\tau = f_v l\sin\theta - f_H l\cos\theta = J\ddot{\theta}$.

Car.



horizon : $u - f_H - M\cdot\ddot{x}$.

vertical : $f_v + Mg - N = 0$.

联立可分

$(M+m)\ddot{x} + ml\cos\theta\,\ddot{\theta} - ml\sin\theta\,\dot{\theta}^2 = U$.

$(J+ml^2)\ddot{\theta} + ml\cos\theta\,\ddot{x} - mgl\sin\theta = 0$

---

③ Lagrange Formulation.

广义坐标. $\vec{q} = [\vec{r}, \vec{\alpha}]^T$ eg. $\vec{q} = \begin{bmatrix} x \\ \theta \end{bmatrix}$.

Lagrange Equation:

$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = Q_j$, $j = 1, 2, \cdots$

T: kinetic energy (动能) — translation / rotation.

V: potential energy (势能) — gravity / spring.

Q_j: generalize force 广义力

virtual displacement (虚位移)

$\delta\vec{r} = \sum_{j=1}^{n} \frac{\partial \vec{r}}{\partial q_j}\,\delta q_j$.

$\delta\vec{\alpha} = \sum_{j=1}^{n} \vec{w}_j\,\delta q_j$.

---

virtual works 虚功.

$\delta W = \sum \vec{F}_n \cdot \delta\vec{r}_n + \sum \vec{\tau}_k \cdot \delta\vec{\alpha}_k$.

$= \sum Q_j\,q_j$.

Eg. $\vec{q} = [x, \theta]^T = [q_1, q_2]^T$

动能 $T = \frac{1}{2}MV_0^2 + \frac{1}{2}mV_A^2 + \frac{1}{2}J\dot{\theta}^2$.

势能 $V = mgl\cos\theta$.

$\vec{v}_0 = \dot{x}\vec{i}$

$\vec{v}_A = \vec{v}_0 + \vec{\omega}_A \times \vec{r}_{OA}$.

$= [\dot{x} + (\dot{\theta}\cos\theta)]\vec{i} - l\dot{\theta}\sin\theta\,\vec{j}$.

$T = \frac{1}{2}M\vec{v}_0^T\vec{v}_0 + \frac{1}{2}m\vec{v}_A^T\vec{v}_A + \frac{1}{2}J\dot{\theta}^2$.

$= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + ml\dot{x}\dot{\theta}\cos\theta + \frac{1}{2}J\dot{\theta}^2$.

虚位移 (外力作用点.) ③

$\delta\vec{r}_0 = \frac{\partial\vec{r}_0}{\partial q_1}\delta q_1 + \frac{\partial\vec{r}_0}{\partial q_2}\delta q_2$.

$= \frac{\partial\vec{r}_0}{\partial x}\delta x + 0\,\delta\theta$.

$= \delta x \cdot \vec{i}$.

虚功 $= \delta W = \vec{U}\cdot\delta\vec{r}_0 = (U\vec{i})(\delta x\,\vec{i})$

$= U\delta x = \sum_{j=1}^{2} Q_j q_j$.

$= Q_1 q_1 + Q_2 q_2$. 广义力.

$\Rightarrow \begin{cases} Q_1 = U \\ Q_2 = 0 \end{cases}$

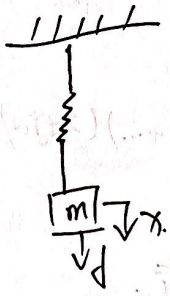
$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = Q_j$.

$\begin{cases} j=1,\ q_1 = x \Rightarrow (M+m)\ddot{x} + m(\cos\theta\cdot\ddot{\theta} - ml\sin\theta\cdot\dot{\theta}^2 = U. \\ j=2,\ q_2 = \theta \Rightarrow (J+ml^2)\ddot{\theta} + ml\cos\theta\,\ddot{x} - mgl\sin\theta = 0 \end{cases}$

↓ 线性化.

$\begin{cases} (M+m)\ddot{x} + ml\ddot{\theta} = U. \\ (J+ml^2)\ddot{\theta} + ml\ddot{x} - mgl\theta = 0 \end{cases}$

Response. Dynamic equation



$m\ddot{x} + kx = f$.

$\mathcal{L}T \Rightarrow$

$m[s^2X(s) - sx(0) - \dot{x}(0)] + kX(s) = F(s)$

$\Rightarrow X(s) = \underbrace{\frac{\dot{x}(0)}{s^2+\frac{k}{m}} + \frac{sx(0)}{s^2+\frac{k}{m}}}_{\text{initial condition}} + \underbrace{\frac{F(s)}{ms^2+k}}_{\text{input}}$.

1) Free response, $f = 0$, $x(0) \neq 0$, $\dot{x}(0) \neq 0$

$\xrightarrow{\text{Inv } \mathcal{L}.T.}$

$X(s) = \frac{\dot{x}(0)}{s^2+\frac{k}{m}} + \frac{sx(0)}{s^2+\frac{k}{m}}$.

$x(t) = A\sin(\sqrt{\frac{k}{m}}\,t + \phi)$ $\omega_n = \sqrt{\frac{k}{m}}$. natural frequency

2) Forced response. $f \neq 0$, $x(0) = \dot{x}(0) = 0$ $f = P\sin\omega t$

$X(s) = \frac{P}{m}\frac{\omega^2}{s^2+\omega^2} \cdot \frac{1}{s^2+\frac{k}{m}}$.

$= \frac{-P\omega\sqrt{\frac{m}{k}}}{k-m\omega^2}\cdot\frac{\sqrt{\frac{k}{m}}}{s^2+\frac{k}{m}} + \frac{P}{k-m\omega^2}\cdot\frac{\omega}{s^2+\omega^2}$.

含衰减

$x(t) = \underbrace{\frac{-P\omega\sqrt{\frac{m}{k}}}{k-m\omega^2}\sin\sqrt{\frac{k}{m}}\,t}_{\text{含衰减}} + \frac{P}{k-m\omega^2}\sin\omega t$.

$x(t)|_{t>t_c} = \frac{P}{k-mw^2}\sin wt$

当 $k-mw^2=0$, $w=w_n=\sqrt{\frac{k}{m}}$ 时

$x(t)\to\infty$ 共振.

$x(t) = \frac{1}{k-mw^2}P\sin wt$

Frequency response.

$X(t)=X\sin wt$ → [$G(s)$] → $y_{ss}(t)$

$G(s)=\frac{Y(s)}{X(s)}$

Impulse response $X(s)=1 \Rightarrow G(s)=Y(s)$

Step response $X(s)=\frac{1}{s} \Rightarrow G(s)=sY(s)$

Frequency response.

$Y(s)=\frac{Xw}{s^2+w^2}=\frac{Xw}{(s+jw)(s-jw)}$.

$G(s)=\frac{P(s)}{q(s)}=\frac{P(s)}{(s+s_1)(s+s_2)+\cdots}$

$\Rightarrow Y(s)=X(s)G(s)=\frac{a}{s+jw}+\frac{\bar{a}}{s-jw}+\frac{b_1}{s+s_1}+\frac{b_2}{s+s_2}+\cdots$

Inv L.T.

$y(t)=ae^{-jwt}+\bar{a}e^{jwt}+b_1e^{-s_1t}+b_2e^{-s_2t}+\cdots$

$t\to\infty$ 时. $y_{ss}(t)=ae^{-jwt}+\bar{a}e^{jwt}$.

Find a and $\bar{a}$.

$Y(s)(s+jw)=G(s)\frac{Xw}{(s+jw)(s-jw)}(s+jw)$

$=a+(\frac{\bar{a}}{s-jw}+\frac{b_1}{s+s_1}+\frac{b_2}{s+s_2}+\cdots)(s+jw)$

when $s=-jw$ 时.

$G(-jw)\frac{Xw}{-2jw}=a=-G(-jw)\frac{X}{2j}$.

同理可得 $\bar{a}=G(jw)\frac{X}{2j}$.

$\Rightarrow y_{ss}(t)=X\frac{G(-jw)e^{-jwt}-G(jw)e^{jwt}}{-2j}$.

$=X|G(jw)|\sin(wt+\phi)$.

Electrical System.

Power = $e\times i$.

~~electron~~

electrical charge. $q(t)=\int_0^t i(t)dt$.

magnetic flux linkage

$\lambda(t)=\int_0^t e(t)dt$, .

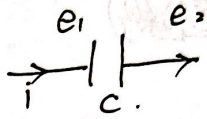
$\Delta W=i(e\,dt)=id\lambda$

$=e(i\,dt)=edq$.

$=ie\,dt$.

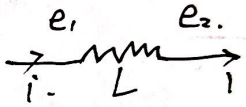
Ideal Elements.

Capacitor



$i(t)=C\frac{d}{dt}(e_1-e_2)$

Inductor



$i(t)=\frac{1}{L}\int_0^t(e_1-e_2)dt$.

Resistor



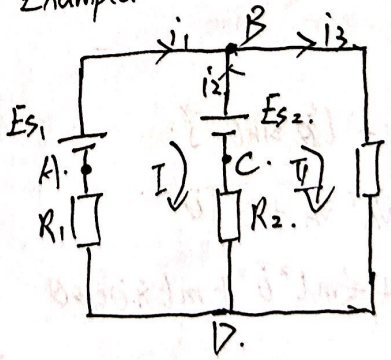
$e_1-e_2=iR_2$.

Physical Laws.

1> Voltage Law $\Sigma e=0$.

2> Current Law $\sum_{node} i=0$.

Example



Current Law: node B $\Sigma i=i_1+i_2-i_3=0$

Voltage Law:

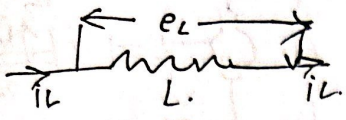
Loop I $\Sigma e=e_{AB}+e_{BC}+e_{CD}+e_{DA}$

$=E_{s_1}-E_{s_2}+i_2R_2-i_1R_1=0$

Loop① $\sum e = e_{CB} + e_{BD} + e_{DC}$

$= E_{s2} - i_3 R_3 - i_2 R_2 = 0$

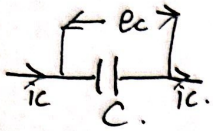
Transfer function (zero ~~con~~initial condition).



$e_L(t) = L \frac{di_L(t)}{dt}$

$E_L(s) = L s I_L(s)$.

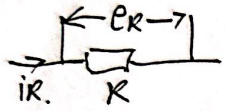
$\frac{E_L(s)}{I_L(s)} = \cancel{Z(s)} \cdot Ls$.



$i_c(t) = C \frac{de_c(t)}{dt}$

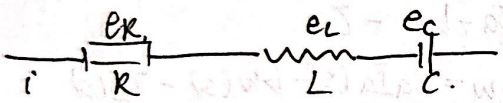
$I_c(s) = Cs E_c(s)$.

$\Rightarrow Z(s) = \frac{E_c(s)}{I_c(s)} = \frac{1}{Cs}$



$Z(s) = \frac{E_R(s)}{I_R(s)} = R$.

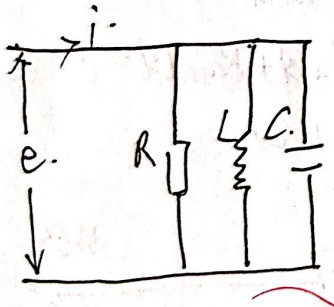
Series.



$Z(s) = \frac{E_R + E_L + E_C}{I} = Z_R + Z_L + Z_C$

$= Ls + R + \frac{1}{Cs}$

Parallel



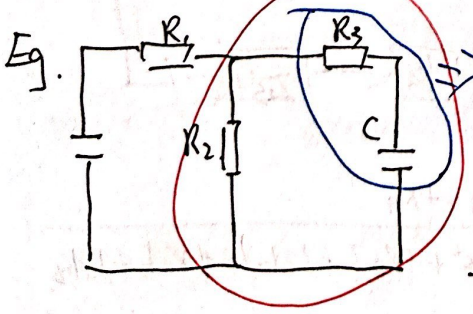
$Z(s) = \frac{E}{I}$

$I = \frac{E}{Z_{R_1}} + \frac{E}{Z_L} + \frac{E}{Z_C}$

$\Downarrow$

$Z(s) = \frac{E}{I} = \frac{1}{\frac{1}{Z_R} + \frac{1}{Z_L} + \frac{1}{Z_C}}$

Eg.



$\Rightarrow Z = Z_{R_3} + Z_C$

$= R_3 + Cs$.

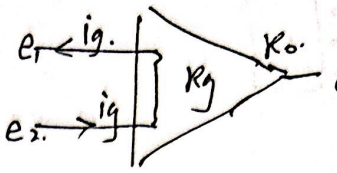
$\frac{1}{Z_1} = \frac{1}{R_2} + \frac{1}{Z}$

$Z_2 = Z_{R_1} + Z_1$

$= \frac{E}{I}$.

~~$Z_2 + Z$~~

Electrical Operational Amplifier. ④

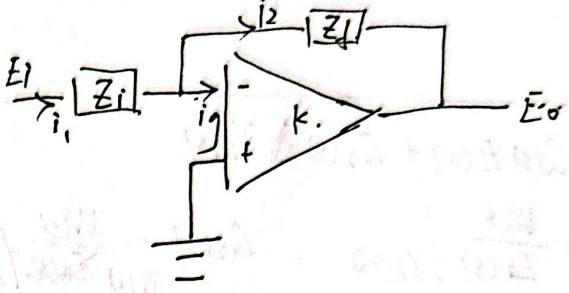


$i_g = 0$. ①

$e_o = k(e_2 - e_1)$ 其中 $k \approx 10^6 \sim 10^8$

$\Rightarrow e_1 = e_2$ ②.

Feedback



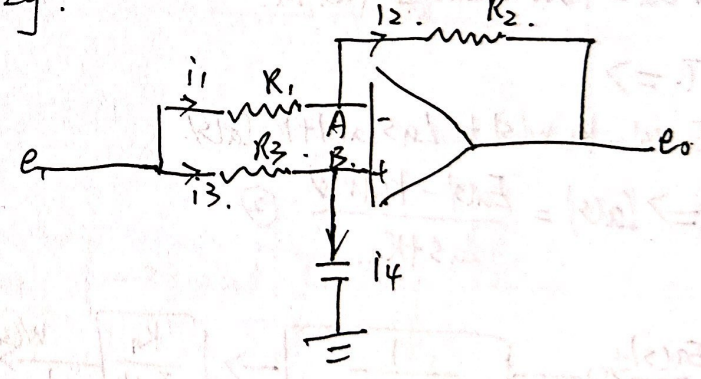
$i_g = 0 \Rightarrow i_1 = i_2$.

$e_g = 0$.

$I_1 = \frac{E_i - 0}{Z_i} = I_2 = \frac{0 - E_o}{Z_f}$.

$\Rightarrow \frac{E_o}{E_i} = -\frac{Z_f}{Z_i}$

Eg.



$E_A = E_B$ $I_1 = I_3$ $I_3 = I_4$

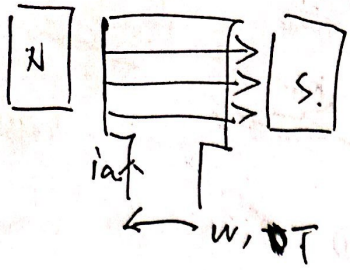
$I_1 = \frac{E_1 - E_A}{R_1} = I_2 = \frac{E_A - E_o}{R_2}$

$I_3 = \frac{E_1 - E_B}{R_3} = I_4 = \frac{E_B - 0}{\frac{1}{Cs}}$

$\Downarrow$

$\frac{E_o}{E_2} = \frac{R_1 + R_2}{R_1(1 + R_2 Cs)} - \frac{R_2}{R_1}$

Motor / Generator.

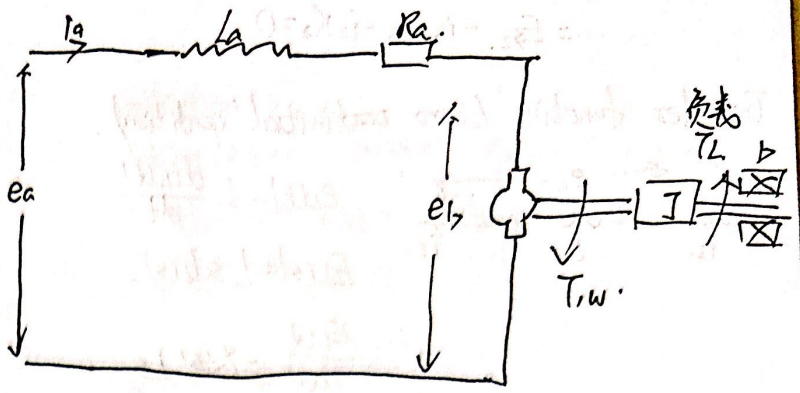


$T = k_a i_a$.

$e_b = k_b w$.

能动势

ideal: $k_a = k_b$



$e_a - L_a \frac{di_a}{dt} - R_a i_a - e_b = 0$.

$J\dot{w} = T - \tau_b - \tau_b = T - bw - \tau_L$.

$W(s) = G_1(s) E_a(s) + G_2(s) T_L(s)$.

$G_1(s) = \frac{W(s)}{E_a(s)}\Big|_{\tau_L=0}$, $G_2(s) = \frac{W(s)}{T_L(s)}\Big|_{E_a=0}$.

(1) when $\tau_L = 0$, Find $\frac{W(s)}{E_a(s)}$

$J\dot{w} = T - bw = k_a i_a - bw$.

L.T. $\Rightarrow (Js+b) W(s) = k_a I_a(s)$.

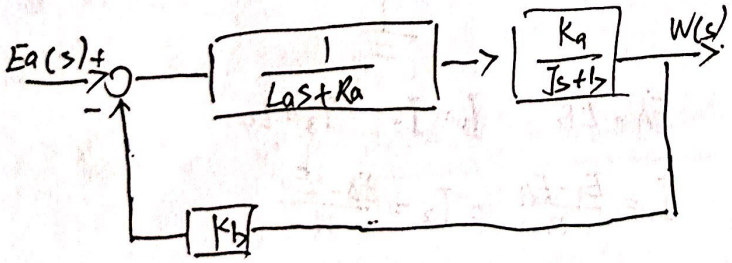
$\Rightarrow \frac{W(s)}{I_a(s)} = \frac{k_a}{Js+b}$ ①.

$e_a = k_b w + L_a \frac{di_a}{dt} + R_a i_a$.

L.T. $\Rightarrow$

$E_a(s) = k_b W(s) + L_a s I_a(s) + R_a I_a(s)$.

$\Rightarrow I_a(s) = \frac{E_a(s) - k_b W(s)}{L_a s + R_a}$ ②.



$\Downarrow$

$$\frac{W(s)}{E_a(s)} = \frac{k_a}{L_a J s^2 + (R_a J + L_a b) s + R_a b + k_a k_b}$$

(2) $e_a = 0$ Find $\frac{W(s)}{T_L(s)}$

Mechanical

$J\dot{w} = T - bw - \tau_L$.

$= k_a i_a - bw - \tau_L$.

L.T $\Rightarrow Js W = k_a I_a(s) - bW(s) - T_L(s)$

$W(s) = \frac{1}{Js+b} [k_a I_a(s) - T_L(s)]$.

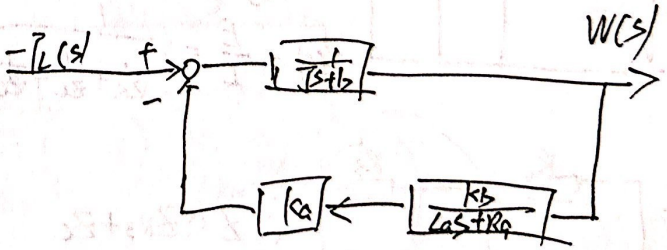
Electrical

$e_b + L_a \frac{di_a}{dt} + R_a i_a = 0$.

$k_b w = L_a \frac{di_a}{dt} + R_a i_a = 0$

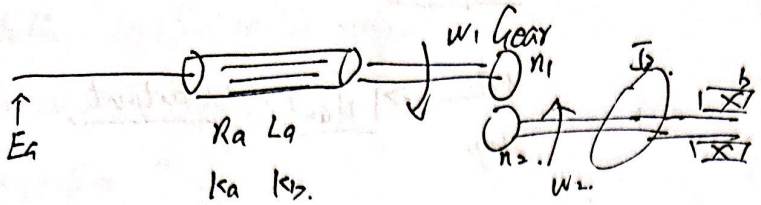
L.T $\Rightarrow k_b W(s) + L_a s I_a(s) + R_a I_a(s) = 0$

$\Rightarrow I_a(s) = \frac{-k_b}{L_a s + R_a} W(s)$.

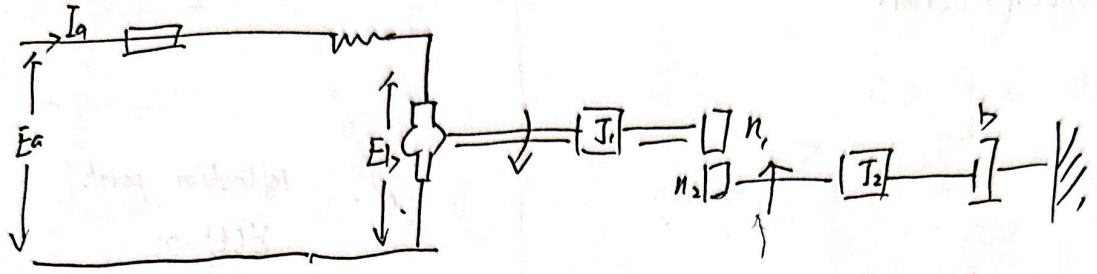


$$\frac{W(s)}{T_L(s)} = \frac{L_a s + R_a}{L_a J s^2 + (R_a J + L_a b) s + R_a b + k_a k_b}$$

Eg.



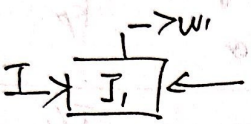
Find $\frac{W_2(s)}{E_a(s)}$.



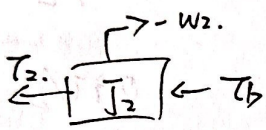
Electrical: $e_a - i_a R_a - L_a \frac{di_a}{dt} - e_b = 0$

Motor $T = k_a i_a$ $\quad e_b = k_b w_1$

Mechanical



$T - \tau_1 = J_1 \dot{w}_1$



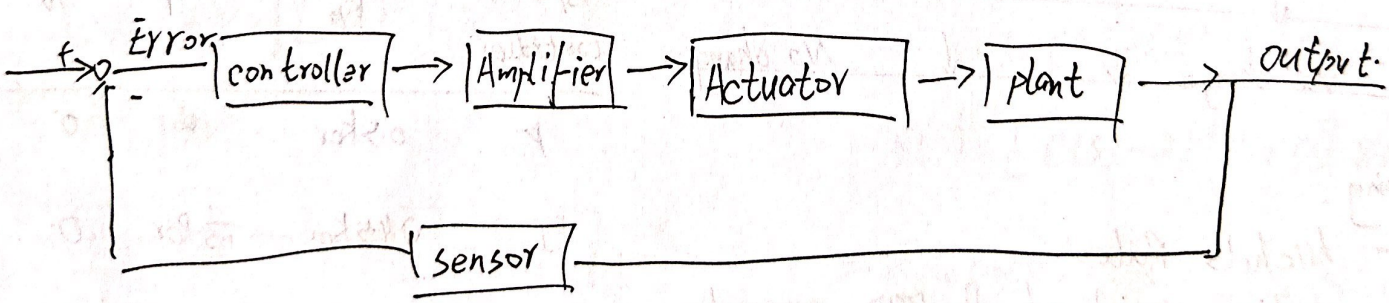
$J_2 \dot{w}_2 = \tau_2 - b w_2$

$\frac{\tau_1}{\tau_2} = \frac{n_1}{n_2} = \frac{w_2}{w_1} = A$

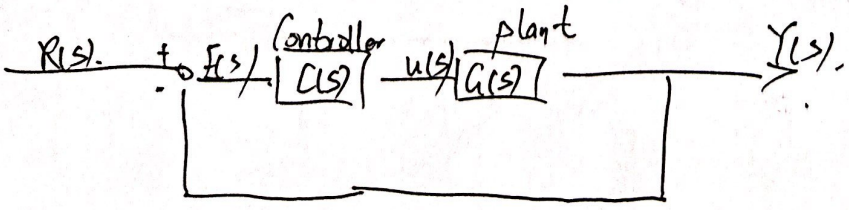
$$\Rightarrow \frac{W_2(s)}{E_a(s)} = \frac{k_a A}{(J_1 + J_2 A^2) L_a s^2 + (J_1 R_a + J_2 A^2 R_a + \cancel{L_a k} L_a b A^2) s + R_a b A^2 + k_a k_b A}$$

PID Control and PID Tuning.

Feedback Control



Goal: drive error to zero



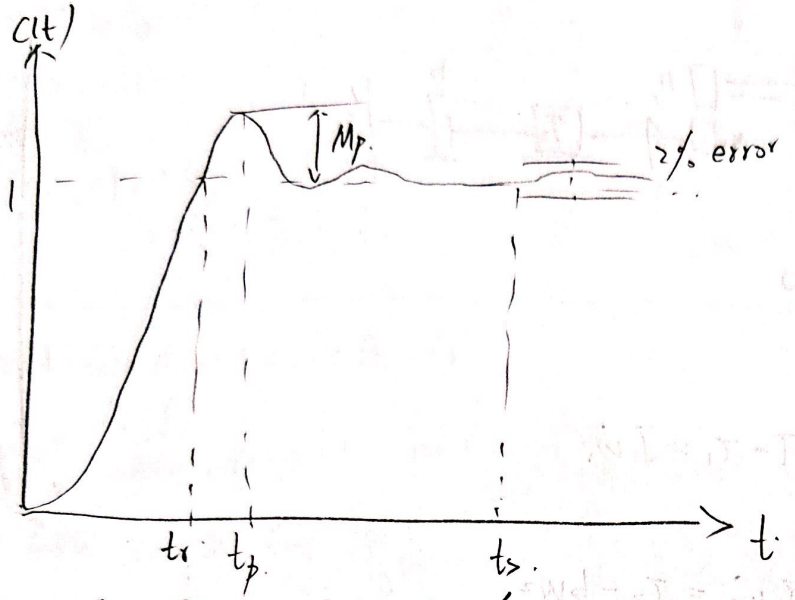
PID: $u(t) = k_p e(t) + k_i \int e(t) dt + k_d \frac{de(t)}{dt}$

$$C(s) = \frac{U(s)}{E(s)} = k_p + \frac{k_i}{s} + k_d s.$$

$$= k_p\left(1 + \frac{1}{T_i s} + T_d s\right).$$

其中 $T_i = \frac{k_p}{k_i}$ $T_d = \frac{k_d}{k_p}$.

Transient response specifications.



t_r: rise time, 0 -> 100%.

t_p: peak time 0 -> first peak.

M_p: maximun overshoot

t_s: setting time 0 -> steady-state.

Response	t_r/t_p	M_p	t_s	S-S error
k_p	↓	↑	Small change.	↓
k_i	↓	↑	↑	Eliminate
k_d	Small change	↓	↓	No change.

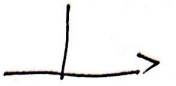
PID Tuning.

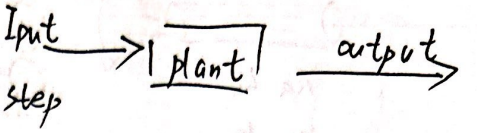
Ziegler-Nichols Rules.

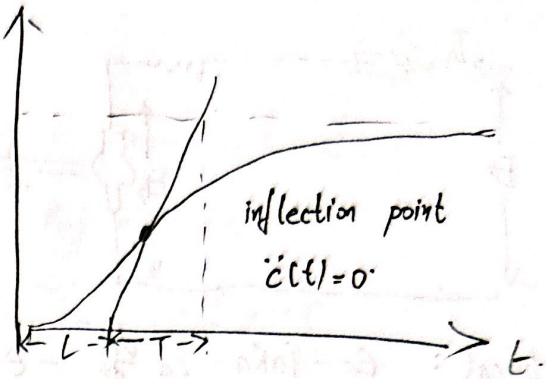
provide a starting point of the PID parameters. 封闭感.

1> First Method: 1st order or over-dampered









Types of Controller	k_p	T_i	T_d
P	T/L	∞	0
PI	$0.9T/L$	$\frac{L}{0.3}$	0
PID	$1.2T/L$	$2L$	$0.5L$

2> Second Method.

a> set $T_i = \infty$, $T_d = 0$, use only k_p

b> increase k_p until c(t) exhibits 临界稳定等幅值振荡 $k_{cr} = k_p$

Type of Controller	k_p	T_i	T_d
P	$0.5k_{cr}$	∞	0
PI	$0.45k_{cr}$	$\frac{1}{1.2}P_{cr}$	0
PID	$0.6k_{cr}$	$0.5P_{cr}$	$0.125P_{cr}$

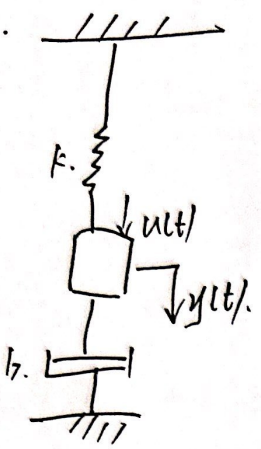
Modern control:

— Multi input, multi output

— Group of ODEs.

— Controller ?

Eg.



Dynamic Equation

$m\ddot{y} + b\dot{y} + ky = u.$

State variable 状态变量

$$\vec{x} = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} y(t) \\ \dot{y}(t) \end{bmatrix}$$

$\dot{x}_1 = \dot{y} = x_2.$

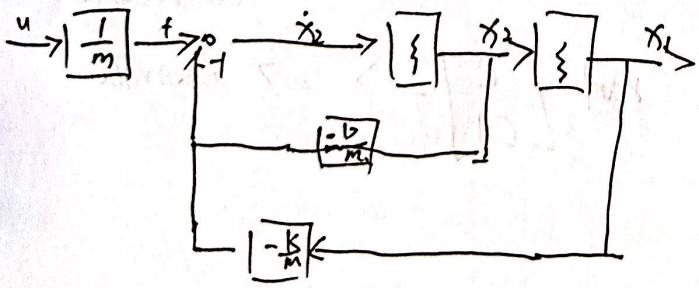
$\dot{x}_2 = \ddot{y} = \frac{k}{m}y - \frac{b}{m}\dot{y} + \frac{u}{m}.$

$= -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{u}{m}.$

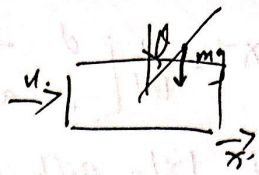
$\Rightarrow$

$$\dot{\vec{x}} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u.$$

$\dot{\vec{x}} = A\vec{x} + Bu.$



Eg.



$(M+m)\ddot{x} + ml\ddot{\theta} = u.$

$(I+ml^2)\ddot{\theta} + ml\ddot{x} - mgl\theta = 0.$

$\Rightarrow a\ddot{x} + b\ddot{\theta} = u$

$c\ddot{\theta} + b\ddot{x} - d\theta = 0$

$$\Rightarrow \begin{cases} \ddot{\theta} = \frac{ad}{ac-b^2}\theta - \frac{b}{ac-b^2}u = \bar{a}\theta - \bar{b}u \\ \ddot{x} = -\frac{abd}{a^2c-ab^2}\theta + \frac{ac}{a^2c-ab^2}u = -\bar{c}\theta + \bar{d}u. \end{cases}$$

~~Take state space~~ $\vec{x} \to A$

let state variable.

$$\vec{x} = \begin{bmatrix} \theta \\ \dot{\theta} \\ x \\ \dot{x} \end{bmatrix} \quad \dot{\vec{x}} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \dot{x} \\ \ddot{x} \end{bmatrix} = \overset{A}{\begin{bmatrix} 0 & 1 & 0 & 0 \\ \bar{a} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -\bar{c} & 0 & 0 & 0 \end{bmatrix}}\vec{x} + \overset{B}{\begin{bmatrix} 0 \\ -\bar{b} \\ 0 \\ \bar{d} \end{bmatrix}} u.$$

$\Downarrow$

State space $\quad \dot{\vec{x}} = A\vec{x} + Bu$

$\vec{y} = C\vec{x} + Du.$

L.T. $\Rightarrow \quad sX(s) = AX(s) + BU(s)$

$Y(s) = CX(s) + DU(s)$

$\Rightarrow X(s) = (sI - A)^{-1}BU(s).$

$Y(s) = [C(sI-A)^{-1}B + D]U(s).$

$$\Rightarrow G(s) = \frac{Y(s)}{U(s)} = C(sI-A)^{-1}B + D.$$

$X=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $X^{-1}=\frac{1}{|X|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ $|X|=$ det $ad-bc$.

$X=\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ $|X|=a_1(b_2c_3-c_2b_3)-a_2(b_1c_3-c_1b_3)+a_3(b_1c_2-c_1b_2)$.

$X^{-1}=\frac{1}{|X|}\begin{bmatrix} b_2c_3-c_2b_3 & c_1b_3-b_1c_3 & b_1c_2-c_1b_2 \\ c_2a_3-a_2c_3 & a_1c_3-c_1a_3 & a_2c_1-a_1c_2 \\ a_2b_3-b_2a_3 & b_1a_3-a_1b_3 & a_1b_2-a_2b_1 \end{bmatrix}$

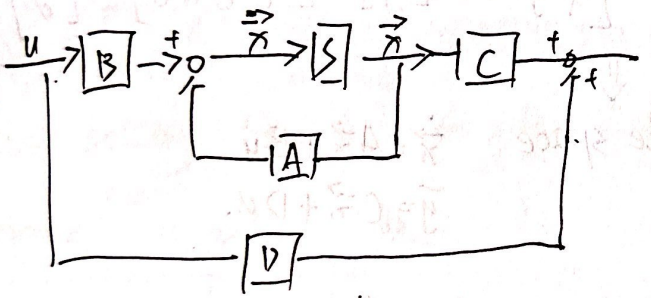
$G(s)=\frac{N(s)}{D(s)}=\frac{N(s)}{|sI-A|}$

① $|sI-A|$ 是特征多项式

② 特征值为 极点

Controllability (可控性)

state space $\begin{cases} \dot{\vec{x}}=A\vec{x}+Bu \\ y=C\vec{x}+Du. \end{cases}$



$\vec{x}$: state variable. (n) A nxn.

$\vec{u}$ input. (r). B nx1

$\vec{y}$ output (m). C mxn

D mxr.

Controllability $u(t) \Rightarrow x(t_0) \xrightarrow{t_0} $ any other $\vec{x}(t)$.

Observability. $y(t) \to x(t)$ $t<\infty$.

Complete state controllable.

rank $[B : AB : \cdots : A^{n-1}B]=n$.

秩

行列式 $|\ |=0$ 不满秩

$|\ |\neq 0$ 满秩.

Output Controllable

rank $[CB : CAB : CA^2B : \cdots : CA^{n-1}B : D]=m$

Observable.

rank $\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}=n$.

Eg. $\dot{\vec{x}}=\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix}=\begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}\vec{x}+\begin{bmatrix} B \\ 0 \\ \vdots \end{bmatrix}u$. (A)

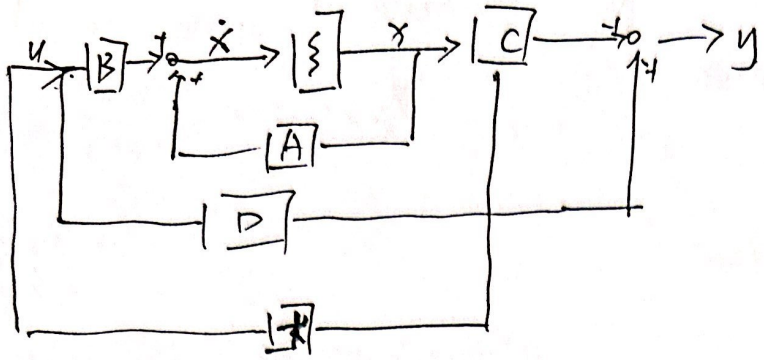
$y=\underbrace{[1,\ 0]}_{C}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $D=0$.

rank $[B : AB]=2$ complete state controllable.

rank $\begin{bmatrix} C \\ CA \end{bmatrix}=2 \Rightarrow$ observable.

Pole Placement Method 极点配置法

$$\begin{cases} \dot{\vec{x}} = A\vec{x} + Bu \\ y = C\vec{x} + Bu \end{cases}$$



$u = -k\vec{x}$ $\quad k = [k_1, k_2 \cdots k_n]$.

$\dot{\vec{x}} = A\vec{x} + Bu = A\vec{x} - Bk\vec{x} = (A - Bk)\vec{x} = A^*\vec{x}$.

eigenvalues of $A^* = A - Bk \Longleftrightarrow$ poles

$|sI - A^*| = |sI - A + Bk| = (s-u_1)(s-u_2)\cdots(s-u_n)$.

$u_1 \cdots u_n$: desired poles

流程:

step 1 : check the controllability.

step 2. derive the characteristic equation

$(s-u_1)(s-u_2)\cdots(s-u_n) = s^n + \alpha_1 s^{n-1} + \cdots + \alpha_n$.

step 3 : let $|sI - A + Bk| =$ → solution of $k = [k_1, \cdots k_n]$

Eg. $\dot{\vec{x}} = A\vec{x} + Bu$. $\quad A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix}$ $\quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

place the desired poles at

$s_1 = -2+4j$ $\quad s_2 = -2-4j$ $\quad s_3 = -10$

step 1 $\quad \text{rank}\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -6 \\ 1 & -6 & 31 \end{bmatrix} = 3$ 满秩.

step 2 $\quad (s-s_1)(s-s_2)(s-s_3) = s^3 + 14s^2 + 60s + 200$

step 3 $\quad |sI - A + Bk| = s^3 + (6+k_3)s^2 + (5+k_2)s + (1+k_1)$

$$\Rightarrow \begin{cases} 6 + k_3 = 14 \\ 5 + k_2 = 60 \\ 1 + k_1 = 200 \end{cases} \Rightarrow k = [k_1, k_2, k_3] = [199, 55, 8].$$

control

$u = -k\vec{x} = -199x_1 - 55x_2 - 8x_3$